

PHYS 3316 — Quantum Mechanics

Section 13721

Notebook 0

Probability, Expectation Values, and the Schrödinger Wave Function

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Learning Objectives

After completing this notebook, you should be able to:

- Normalize one-dimensional wave functions.
 - Compute probabilities and expectation values.
 - Interpret probability densities.
 - Verify analytical results numerically.
 - Visualize quantum wave functions.
 - Apply Ehrenfest's theorem to the harmonic oscillator.
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Required Reading

Griffiths & Schroeter *Introduction to Quantum Mechanics (3rd Ed.)*

- Chapter 1
 - Sections 1.1–1.5
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Software

This notebook uses

- Python 3
 - NumPy
 - SciPy
 - Matplotlib
-

Submission Instructions

Submit a single PDF generated from this notebook.

Your notebook should contain

- completed code cells,
- written derivations,
- figures,
- discussion questions,
- and all requested calculations.

Name your file

NB0_LastName_StudentID.pdf

Honor Statement

You are encouraged to discuss the assignment with your classmates. However, all submitted work must represent your own understanding. You should be able to explain every derivation, calculation, and line of code appearing in your notebook.

```
In [ ]: # =====
# PHYS 3316 – Quantum Mechanics
# Notebook Initialization
# =====

import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import quad
from scipy.linalg import eigh

plt.rcParams["figure.figsize"] = (7,5)
plt.rcParams["font.size"] = 12
plt.rcParams["axes.grid"] = True
```

In these problems, I will give you pre-defined quantum wavefunctions, generally writing a Ψ or ψ that may correspond to physical systems. For now, we're learning how to interpret and understand Ψ . For notation, I tend to denote Ψ as a time-evolving solution and ψ as stationary solution.

Problem 1 — Normalization, Probability, and Expectation Values

Consider the one-dimensional wave function

$$\Psi(x, 0) = \begin{cases} A \frac{x}{a}, & 0 \leq x \leq a, \\ A \frac{b-x}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

where ($0 < a < b$), and (A) is a real normalization constant.

(a) Normalization

Use the normalization condition

$$\int_{-\infty}^{\infty} |\Psi(x, 0)|^2 dx = 1$$

to determine (A) in terms of (a) and (b). Show all steps in your calculation.

(b) Wave function and probability density

Sketch the wave function ($\Psi(x, 0)$) and the corresponding probability density

$$\rho(x) = |\Psi(x, 0)|^2.$$

Clearly indicate the points ($x=0$), ($x=a$), and ($x=b$).

(c) Most probable position

At what position is the particle most likely to be found?

Explain how this position is determined from the probability density.

(d) Probability in the interval ($0 \leq x \leq a$)

Calculate the probability that a measurement of the particle's position gives a value in the interval

$$0 \leq x \leq a.$$

That is, evaluate

$$P(0 \leq x \leq a) = \int_0^a |\Psi(x, 0)|^2 dx.$$

(e) Expectation value of position

Calculate the expectation value

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, 0)|^2 dx.$$

Your final answer should be expressed in terms of (a) and (b).

(f) Interpretation

Compare the expectation value ($\langle x \rangle$) with the most probable position.

Are the expectation value and the most probable value necessarily the same? Explain briefly.

YOUR ANSWER HERE

Problem 2 — An Exponentially Localized Wave Function

Consider the wave function

$$\Psi(x, t) = Ae^{-\lambda|x|}e^{-i\omega t},$$

where ($\lambda > 0$), (ω) is a real constant, and (A) is a real normalization constant.

(a) Normalization

Use the normalization condition

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

to determine (A) in terms of (λ). You may use the symmetry of the probability density to write

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 2 \int_0^{\infty} |\Psi(x, t)|^2 dx.$$

(b) Probability density

Compute the probability density

$$\rho(x, t) = |\Psi(x, t)|^2.$$

Does the probability density depend on time? Explain why or why not.

(c) Expectation value of position

Before evaluating any integrals, use symmetry to predict the value of

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx.$$

Then verify your prediction by direct calculation.

(d) Expectation value of x^2

Calculate

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 |\Psi(x, t)|^2 dx.$$

Express your answer in terms of λ .

(e) Position uncertainty

Use

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

to determine the uncertainty in position.

(f) Probability outside one standard deviation

Calculate the probability that the particle is found outside the interval

$$-\sigma_x \leq x \leq \sigma_x.$$

That is, evaluate

$$P(|x| > \sigma_x) = \int_{-\infty}^{-\sigma_x} |\Psi(x, t)|^2 dx + \int_{\sigma_x}^{\infty} |\Psi(x, t)|^2 dx.$$

Use symmetry to simplify the calculation.

(g) Interpretation

Briefly explain how the parameter λ affects the spatial localization of the particle. What happens to σ_x as λ increases?

YOUR ANSWER HERE:

Problem 3 — Conservation of Probability and Probability Current

The probability of finding a particle between positions a and b at time t is

$$P_{ab}(t) = \int_a^b |\Psi(x, t)|^2 dx.$$

The time-dependent Schrödinger equation is

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x, t)\Psi.$$

Its complex conjugate is

$$-i\hbar \frac{\partial \Psi^*}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + V(x, t)\Psi^*,$$

where the potential $V(x, t)$ is assumed to be real.

(a) Time derivative of the probability density

Starting from

$$\rho(x, t) = |\Psi(x, t)|^2 = \Psi^*(x, t)\Psi(x, t),$$

show that

$$\frac{\partial \rho}{\partial t} = \Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t}.$$

Use the Schrödinger equation and its complex conjugate to eliminate the time derivatives.

(b) Continuity equation

Show that your result can be written in the form

$$\frac{\partial \rho}{\partial t} + \frac{\partial J}{\partial x} = 0,$$

where

$$J(x, t) = \frac{\hbar}{2mi} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right).$$

This equation is called the probability continuity equation.

(c) Probability in a finite interval

Integrate the continuity equation from $x = a$ to $x = b$ and show that

$$\frac{dP_{ab}}{dt} = J(a, t) - J(b, t).$$

Interpret this result in terms of probability flowing into and out of the interval.

(d) Conservation of total probability

Take the limits $a \rightarrow -\infty$ and $b \rightarrow \infty$.

Under what conditions does

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 0$$

follow?

(e) Units and interpretation

Determine the physical units of the probability current $J(x, t)$.

Explain what the sign of J indicates about the direction of probability flow.

(f) Plane wave example

Consider the plane wave

$$\Psi(x, t) = Ae^{i(kx - \omega t)}.$$

Calculate the probability density ρ and probability current J . Show that

$$J = \frac{\hbar k}{m} |\Psi|^2.$$

How is the sign of J related to the sign of k ?

Your answer here:

Problem 4 — Ehrenfest's Theorem for the Harmonic Oscillator

For a particle moving in one dimension, the expectation value of the momentum is

$$\langle p \rangle = \int_{-\infty}^{\infty} \Psi^*(x, t) \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi(x, t) dx.$$

The time-dependent Schrödinger equation is

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x)\Psi.$$

Assume that $\Psi(x, t)$ and its spatial derivatives vanish sufficiently rapidly as $x \rightarrow \pm\infty$.

(a) Time derivative of the momentum expectation value

Differentiate the momentum expectation value with respect to time:

$$\frac{d\langle p \rangle}{dt} = \frac{d}{dt} \int_{-\infty}^{\infty} \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx.$$

Use the product rule to show that

$$\frac{d\langle p \rangle}{dt} = \int_{-\infty}^{\infty} \left[\frac{\partial \Psi^*}{\partial t} \left(-i\hbar \frac{\partial \Psi}{\partial x} \right) + \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \frac{\partial \Psi}{\partial t} \right] dx.$$

(b) Operator form of the equation of motion

The expression obtained above may be written in Dirac notation as

$$\frac{d\langle p \rangle}{dt} = \left\langle \frac{\partial \Psi}{\partial t} \left| \hat{p} \right| \Psi \right\rangle + \left\langle \Psi \left| \hat{p} \right| \frac{\partial \Psi}{\partial t} \right\rangle,$$

where

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}.$$

The time-dependent Schrödinger equation and its adjoint may be written as

$$i\hbar \frac{\partial}{\partial t} |\Psi\rangle = \hat{H} |\Psi\rangle$$

and

$$-i\hbar \frac{\partial}{\partial t} \langle \Psi| = \langle \Psi| \hat{H}.$$

Use these equations to show that

$$\frac{\partial}{\partial t} |\Psi\rangle = -\frac{i}{\hbar} \hat{H} |\Psi\rangle$$

and

$$\frac{\partial}{\partial t} \langle \Psi| = \frac{i}{\hbar} \langle \Psi| \hat{H}.$$

Substitute these results into the time derivative of the momentum expectation value and show that

$$\frac{d\langle p \rangle}{dt} = \frac{i}{\hbar} \langle \Psi | \hat{H} \hat{p} | \Psi \rangle - \frac{i}{\hbar} \langle \Psi | \hat{p} \hat{H} | \Psi \rangle.$$

Using the commutator definition

$$[\hat{H}, \hat{p}] = \hat{H} \hat{p} - \hat{p} \hat{H},$$

deduce that

$$\frac{d\langle p \rangle}{dt} = \frac{i}{\hbar} \langle \Psi | [\hat{H}, \hat{p}] | \Psi \rangle$$

or, equivalently,

$$\frac{d\langle p \rangle}{dt} = \frac{1}{i\hbar} \langle \Psi | [\hat{p}, \hat{H}] | \Psi \rangle.$$

This is the Heisenberg equation of motion for the expectation value of an operator with no explicit time dependence.

For a general operator $\hat{A}(t)$ that may depend explicitly on time, the corresponding equation is

$$\frac{d\langle A \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle.$$

Explain why the final term vanishes for the momentum operator $\hat{p}$ used here.

Instructor's Note: This problem is meant to give you your first exposure to **Dirac notation**. Right now it may seem like we're replacing familiar mathematics with a collection of funny angle brackets. Trust me—we're not making things harder just to make them look more "quantum." Quite the opposite.

Take a few minutes to find Paul Dirac's paper introducing this notation and read the opening paragraph. Dirac developed this formalism because it made calculations easier to think about and easier to organize. If Dirac says it simplifies quantum mechanics...I'm inclined to believe him.

(c) Use of the Schrödinger equation

Use the Schrödinger equation and its complex conjugate to replace the time derivatives $\partial\Psi/\partial t$ and $\partial\Psi^*/\partial t$.

Separate the resulting expression into kinetic-energy and potential-energy terms.

Show that the kinetic-energy terms reduce to boundary terms that vanish under the stated boundary conditions.

(d) Ehrenfest relation for momentum

Show that the remaining potential-energy contribution gives

$$\frac{d\langle p \rangle}{dt} = - \int_{-\infty}^{\infty} \Psi^*(x, t) \frac{\partial V}{\partial x} \Psi(x, t) dx.$$

Therefore,

$$\boxed{\frac{d\langle p \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle}$$

This is Ehrenfest's theorem for the momentum expectation value.

(e) Harmonic oscillator potential

For the one-dimensional harmonic oscillator,

$$V(x) = \frac{1}{2} m \omega^2 x^2.$$

Compute the force associated with this potential,

$$-\frac{\partial V}{\partial x},$$

and use Ehrenfest's theorem to show that

$$\boxed{\frac{d\langle p \rangle}{dt} = -m\omega^2 \langle x \rangle}$$

where

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx.$$

(f) Equation of motion for the position expectation value

Use the Ehrenfest relation

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}$$

to compute the second time derivative of the position expectation value,

$$\frac{d^2 \langle x \rangle}{dt^2}.$$

Show that

$$\boxed{\frac{d^2 \langle x \rangle}{dt^2} + \omega^2 \langle x \rangle = 0}$$

(g) General solution

Write the general solution for the expectation value of position in the form

$$\langle x \rangle(t) = C \cos(\omega t) + D \sin(\omega t),$$

where C and D are determined by the initial values of $\langle x \rangle$ and $\langle p \rangle$.

Express C and D in terms of

$$\langle x \rangle(0) \quad \text{and} \quad \langle p \rangle(0).$$

(h) Physical interpretation

Briefly explain the significance of the result

$$\frac{d^2 \langle x \rangle}{dt^2} = -\omega^2 \langle x \rangle.$$

In your discussion, address the following questions:

1. How does the motion of $\langle x \rangle$ compare with the motion of a classical harmonic oscillator?
2. Does this result imply that the quantum particle follows a definite classical trajectory?
3. What quantity is obeying the classical equation of motion?

Your answers here:

Problem 5 — Numerical Normalization of a Wave Function

Return to the piecewise wave function from Problem 1,

$$\Psi(x, 0) = \begin{cases} A \frac{x}{a}, & 0 \leq x \leq a, \\ A \frac{b-x}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

For this problem, use the parameter values

$$a = 1 \quad \text{and} \quad b = 3.$$

The goal is to determine the normalization constant numerically and compare it with your analytical result from Problem 1.

(a) Define the unnormalized wave function

Define the unnormalized function

$$f(x) = \begin{cases} \frac{x}{a}, & 0 \leq x \leq a, \\ \frac{b-x}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Then write the normalized wave function as

$$\Psi(x, 0) = A f(x).$$

In a Python code cell, define a function named `f(x, a, b)` that returns the appropriate value of the piecewise function.

Your function should return zero when $x < 0$ or $x > b$.

(b) Compute the normalization integral

The normalization condition is

$$\int_{-\infty}^{\infty} |\Psi(x, 0)|^2 dx = 1.$$

Substituting $\Psi(x, 0) = A f(x)$ gives

$$A^2 \int_{-\infty}^{\infty} |f(x)|^2 dx = 1.$$

Because the wave function is nonzero only on the interval $0 \leq x \leq b$, compute the integral

$$I = \int_0^b |f(x)|^2 dx$$

numerically using `scipy.integrate.quad`.

Display the numerical value of I .

(c) Determine the numerical normalization constant

Use the numerical integral to compute the normalization constant

$$A_{\text{num}} = \frac{1}{\sqrt{I}}.$$

Display the value of A_{num} .

(d) Compare with the analytical result

Using your result from Problem 1, evaluate the analytical normalization constant

$$A_{\text{exact}}$$

for $a = 1$ and $b = 3$.

Compute the absolute error,

$$\epsilon_{\text{abs}} = |A_{\text{num}} - A_{\text{exact}}|,$$

and the relative error,

$$\epsilon_{\text{rel}} = \frac{|A_{\text{num}} - A_{\text{exact}}|}{|A_{\text{exact}}|}.$$

Display both errors.

(e) Verify normalization directly

Define the normalized wave function

$$\Psi(x, 0) = A_{\text{num}} f(x).$$

Then compute

$$N = \int_0^b |\Psi(x, 0)|^2 dx$$

numerically.

Your result should satisfy

$$N \approx 1.$$

Report the value of N and comment on the numerical accuracy.

(f) Short interpretation

In two or three sentences, explain why numerical normalization is useful even when an analytical solution is available.

Also explain why the limits of integration can be reduced from $(-\infty, \infty)$ to $(0, b)$ for this wave function.